

Hyungtak Lee

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Ph.D. student specializing in Bayesian Optimization, Gaussian processes, and Preference Learning (Human-in-the-loop)

EDUCATION

Iowa State University

Ames, IA

Ph.D. in Computer Science Advisor: Christopher J. Quinn GPA: 3.96/4.0 Aug. 2023 – Expected May 2029

Kwangwoon University

Seoul, Korea

B.S. in Computer Engineering GPA: 4.29/4.5 (3.81/4.0), Dean's List (3 semesters) Mar. 2017 – Feb. 2023

RESEARCH EXPERIENCE

Ph.D. Researcher | Iowa State University

Aug. 2024 – Present

- Developing **approximate inference methods** for **preferential Gaussian processes**, which learn latent utility functions from **human choice feedback**, with applications to **preference-based Bayesian optimization**
- Reimplemented and extended preferential Bayesian optimization methods in **GPyTorch/PyTorch**, including a migration of prior GPy/NumPy code

Research Project | Uncertainty Quantification for LLMs

2026 – Present

- Developing methods for **Uncertainty quantification**-based fine-tuning for large language models; **co-first author** on the resulting manuscript

Undergraduate Research Assistant | Computer Communications Lab, Kwangwoon Univ.

Jul. 2021 – Aug. 2023

- Proposed a GAN architecture that generates training images together with bounding-box labels to augment scarce object-detection datasets (fire detection); first-author journal paper and registered Korean patent
- Contributed to government-funded projects on generative data augmentation and lightweight edge object detection

PUBLICATIONS & MANUSCRIPTS

- Minoo Hong^{*}, **Hyungtak Lee**^{*}, et al. “Uncertainty Quantification-Based Fine-Tuning for Large Language Models.” *Manuscript in preparation*, 2026. (^{*} equal contribution)
- Hyungtak Lee**, Seongju Kang, and Kwangsue Chung. “Robust Data Augmentation Generative Adversarial Network for Object Detection.” *Sensors*, 2023. [pdf]
- Hyungtak Lee**, Seongju Kang, and Kwangsue Chung. “Object Detection with Dataset Augmentation for Fire Images Based on GAN.” *ICTC*, 2022. [pdf]
- Hyungtak Lee**, Seongju Kang, Jaegi Hwang, and Kwangsue Chung. “Object Detection Dataset Generation Framework using GAN.” *Korea Computer Congress (KCC)*, 2022. [pdf]
- Chi-Hyun Ahn, **Hyungtak Lee**, and Kwangsue Chung. “IoT Collaboration System Based on Edge Computing for Smart Livestock System.” *JKIICE*, 2022. [pdf]
- Patent**: “Generative Adversarial Networks for Fire Data Augmentation,” KR 10-2931826 (reg. Feb. 2026), co-inventor.

TEACHING

Teaching Assistant | Principles of AI (COMS 4720/5720), Iowa State University

Jan. 2026 – Present

- Lead two weekly recitation sections and office hours for a joint undergrad/grad course of about 80 students covering search, CSPs, logic, probabilistic reasoning (Bayesian networks, MCMC, HMMs), and machine learning

Teaching Assistant | Computer Architecture (COMS 3210), Iowa State University

Aug. 2023 – Dec. 2024

- Graded LEGv8 assembly and architecture assignments and held office hours on ISA design, pipelining, and memory hierarchy (3 semesters)

OTHER EXPERIENCE

Korean Military Service (KATUSA) | Assigned to Eighth U.S. Army

Dec. 2019 – Jul. 2021

- Mandatory service as a network specialist in a U.S. Army signal brigade; awarded the Army Commendation Medal

TECHNICAL SKILLS

Methods: Bayesian Optimization, Gaussian Processes, Preference Learning, Bandits, Uncertainty Quantification, PEFT

Languages: Python, C/C++, Rust, ARMv8 Assembly

Frameworks & Tools: PyTorch, BoTorch, GPyTorch, NumPy, CUDA, Git, Linux, L^AT_EX